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% Lab 2 Prelab: Step Response Simulations
clear; clc; close all;

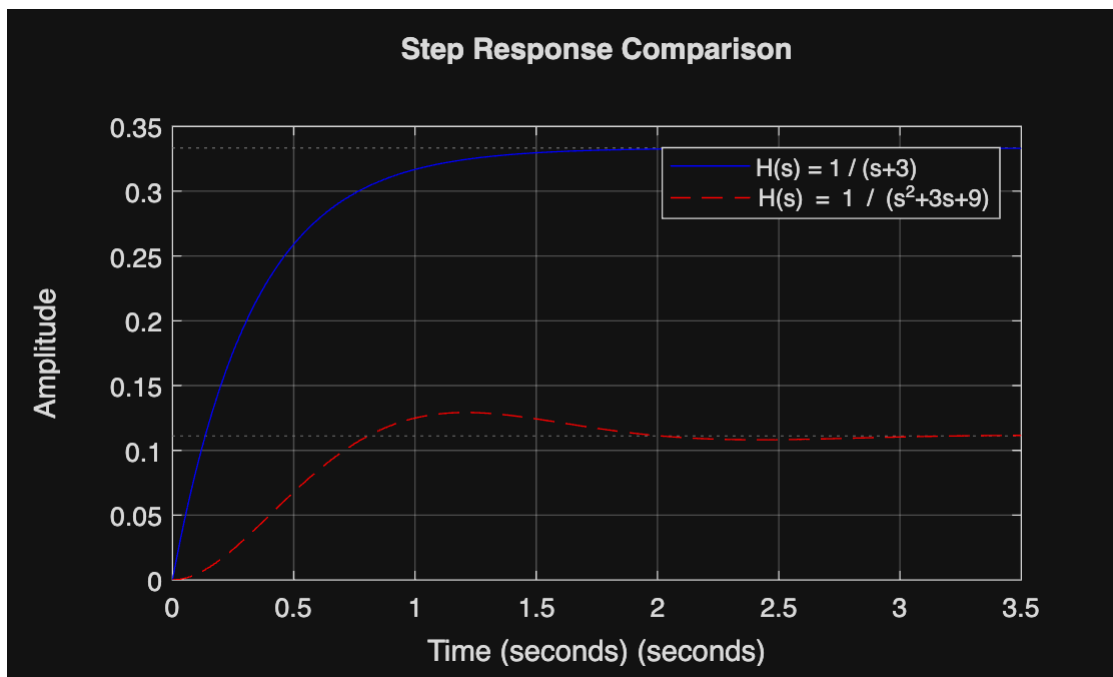
% Define the Laplace variable s
s = tf('s');

% (a) First-order system:  $H(s) = 1 / (s + 3)$ 
H_a = 1 / (s + 3);

% (b) Second-order system:  $H(s) = 1 / (s^2 + 3s + 9)$ 
H_b = 1 / (s^2 + 3*s + 9);

% Create the plot
figure;
step(H_a, 'b-', H_b, 'r--'); % Blue solid for (a), Red dashed for (b)
grid on;
title('Step Response Comparison');
xlabel('Time (seconds)');
ylabel('Amplitude');
legend('H(s) = 1 / (s+3)', 'H(s) = 1 / (s^2+3s+9)');

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% Optional: Display system characteristics in Command Window
disp('Characteristics for System (b):');

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Characteristics for System (b):

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stepinfo(H_b)

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ans = struct with fields:
    RiseTime: 0.5463
    TransientTime: 2.6920
    SettlingTime: 2.6920
    SettlingMin: 0.1035

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SettlingMax: 0.1292
Overshoot: 16.2929
Undershoot: 0
Peak: 0.1292
PeakTime: 1.1973